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# Efficient Treatment and Simultaneous Energy Recovery from Cattle Wastewater by the Micro-Aerated UASB and SBR Coupled System

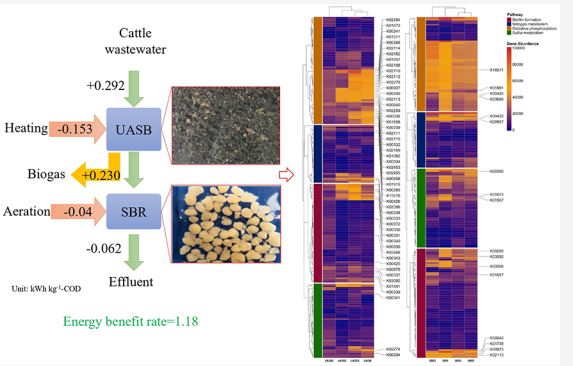
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**ABSTRACT:** The present study introduces a coupled system comprising an upflow anaerobic sludge blanket (UASB) and a sequencing batch reactor (SBR) to treat intensive cattle farming wastewater. The results confirmed that the optimal conditions for UASB were an organic load rate of 9.98 kg-COD $m^{-3}$·$day^{-1}$, a C/N ratio of 22, and a hydraulic retention time (HRT) of 1.5 h. For the SBR reactor, the optimal conditions are a dissolved oxygen (DO) concentration of 6.08 mg $L^{-1}$ and the S2 aeration mode. Under these optimal conditions, the study achieved anaerobic and aerobic sludge granules (0.25−0.5 and 0.5−0.75 mm) with excellent properties due to the cooperation of a functional microbial community, including *Synergistetes*, *Thermotogae*, *Acidobacteria*, *Proteobacteria*, *Chloroflexi*, and *Hyd24−12* in UASB and *Paracoccus*, *Proteobacteria*, *Chloroflexi*, and *Acinetobacter* in SBR. This hybrid system contributed 94% of chemical oxygen demand (COD), 90% of $NH_4^+$-N, 90% of TN, and 61% of TP. Additionally, it achieved 18% net energy recovery from the real cattle wastewater, indicating that the UASB−SBR-coupled system holds potential as an approach for treating and recovering energy from cattle wastewater.




**KEYWORDS:** *cattle wastewater, UASB−SBR-coupled system, anaerobic/aerobic activated sludge granulation, microbial community, energy balance*

## INTRODUCTION

Intensive cattle farming has increasingly been adopted worldwide, resulting in the generation of substantial amounts of wastewater.[1] This wastewater can reach up to 130 L per animal per day and contains high concentrations of pollutants, including $BOD_5$ ranging from 2000 to 30,000 mg $L^{-1}$, total nitrogen ranging from 200 to 2055 mg $L^{-1}$, ammonia between 110 and 1650 mg $L^{-1}$, total phosphorus varying from 100 to 620 mg $L^{-1}$, and suspended solid 200−1200 mg $L^{-1}$.[2] Uncontrolled discharge of the concentrate contaminants can cause DO depletion in water courses, eutrophication, groundwater contamination, and air pollution by ammonia volatilization.[3,4] Consequently, cattle wastewater has become an emerging environmental concern and more robust treatment techniques are required to meet increasingly stringent environmental standards in cattle farming.

Due to the slow degradation of complex pollutants and high nitrogen and phosphorus levels, as well as the occurrence of hydraulic and organic shock loads, specific technology is required to treat cattle wastewater.[5] A coupled system of upflow anaerobic sludge blanket (UASB) and sequencing batch reactor (SBR) with aerobic granular sludge (AGS) has the potential to efficiently treat cattle wastewater. Compared to conventional treatment systems, granular sludge systems have lower reactor volume requirements and lower sludge handling costs while also exhibiting high removal efficiencies of organic matter, nitrogen, and phosphorous.[6] This system is capable of recovering energy from wastewater through biogas production, while simultaneously achieving the removal of organic pollutants and meeting the nitrogen and phosphorus concentration requirements for effluent from the aerobic stage in accordance with surface water discharge standards.[7−9]

However, there are some obstacles that hinder the widespread use of the hybrid UASB and SBR system, such as a long start-up period, low stability, and disintegration.[10,11] In order to effectively capture a majority of the organic matter within a short retention time in the UASB, micro-aeration is employed, which accelerates the hydrolysis of organics (the rate-limiting step of the anaerobic process).[12,13] Proper micro-aeration in the UASB system does not hinder the formation of anaerobic granular sludge (AnGS).[14] On the other hand, the

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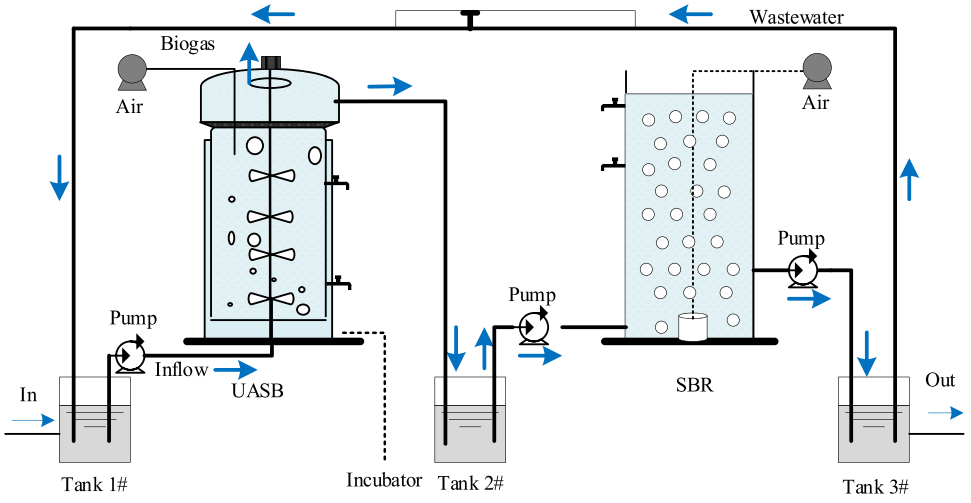



**Figure 1.** Configuration of the UASB−SBR-coupled system.

**Table 1. Cattle Wastewater Properties**

| | COD | $NH_4^+$-N | TN | TP | SS | |
|---|---|---|---|---|---|---|
| cattle wastewater | | | mg $L^{-1}$ | | | pH |
| concentration | 9000−11,800 | 389−651 | 450−700 | 16−28 | 130 ∼ 1200 | 6.8−7.5 |

SBR system presents high potential for nitrogen and phosphorus removal. However, its performance depends on the dissolved oxygen (DO) level since a low DO level cannot assist the formation of activated granular sludge (AGS),[15] while excessive DO can not only affect microbial growth but also be detrimental to energy consumption and phosphorus removal.[16]

To date, the integration of two different granular sludge systems for the efficient removal of high-concentration pollutants has been rarely reported and the conditions for the rapid cultivation of AnGS and AGS have not been disclosed yet. The objective of this study is to explore the proper conditions for rapid granulation of AnGS and AGS to remove pollutants in cattle wastewater. To achieve this goal, two identical UASB reactors with and without micro-aeration are constructed and operated under similar conditions for comparison. The SBR reactor is also optimized by altering the DO concentration and aeration mode. By analyzing the differences in sludge properties, nutrient removal, and microbial communities, we reveal the relevant mechanism of intensive cattle farming wastewater treatment via the hybrid system of UASB and SBR. By addressing real wastewater samples, this study provides valuable insights into the practical application and effectiveness of the combined granular sludge system in the removal of high-concentration pollutants.

## METHODS AND MATERIALS

**UASB−SBR Configuration.** The schematic diagram in Figure 1 illustrates the flow of the test device. Two UASB reactors, each with a water jacket and a three-phase separation zone, were constructed from acrylic material. The entire reactor had a working volume of 7.5 L and was divided into three sections: the bottom section had an inner diameter of 80 mm and a height of 7.52 cm, the middle section had an inner diameter of 160 mm and a height of 40 mm, and the top section had an inner diameter of 160 mm and a height of 210 mm. The SBR, also made of acrylic, had a working volume of 4.27 L, with a height of 1000 mm and an inner diameter of 80 mm. The reactor's top was fitted with an electric stirrer, pH, and DO monitors, while the bottom had a rubber aeration plate connected to an air pump and a gas-flow meter. To adjust the hydraulic retention time and for sampling analysis, a sampling port was installed every 10 cm along the reactor's exterior. The reactor effluent pipe was connected to a solenoid valve controlled by a programmable controller.

**Start-up and Operation.** To initiate the experiment, we inoculated 10.0 g $L^{-1}$ of anaerobic sludge and 3.2 g $L^{-1}$ of aerobic sludge into the UASB and SBR, respectively. Both sludges were collected from the Chengnan wastewater treatment plant in Chongqing, China. We cultivated and optimized the UASB and SBR independently using synthetic wastewater to achieve the best possible performance. We then conducted two different experiments on the UASBs. In the first experiment, we aerated one of the UASB reactors with DO levels around 1 mg $L^{-1}$ at the top to stimulate bacterial growth, as the DO level in the reaction zone was low. In the second experiment, we operated another UASB in strict anaerobic conditions for 120 days (phase I).

The SBR was cultivated with two aeration modes: S1 (5 min inlet → 320 min aeration → 30 min sedimentation → 5 min drainage) and S2 (5 min inlet → 90 min aeration → 60 min aeration- suspending → 60 min aeration → 75 min aeration-suspending → 45 min aeration → 30 min sedimentation → 5 min drainage). Before treating real cattle wastewater, we optimized both the UASB and SBR. The influent COD concentration of the UASB fluctuated irregularly between 3000 and 15,000 mg $L^{-1}$, with an organic load ranging between 2.366 and 12.326 kg COD $m^{-3}$ $day^{-1}$. We controlled the upper DO level at about 1 mg $L^{-1}$, and the C/N ratios were 25:1, 22:1, 20:1, 18:1, 15:1, and 10:1. The HRTs were 12, 20, 24, 36, and 48 h, respectively. The DO concentrations of the SBR were 3.10, 4.12, 5.19, 6.08, 7.11, and 8.01 mg $L^{-1}$, respectively. After reaching the optimal state, we combined the UASB and SBR


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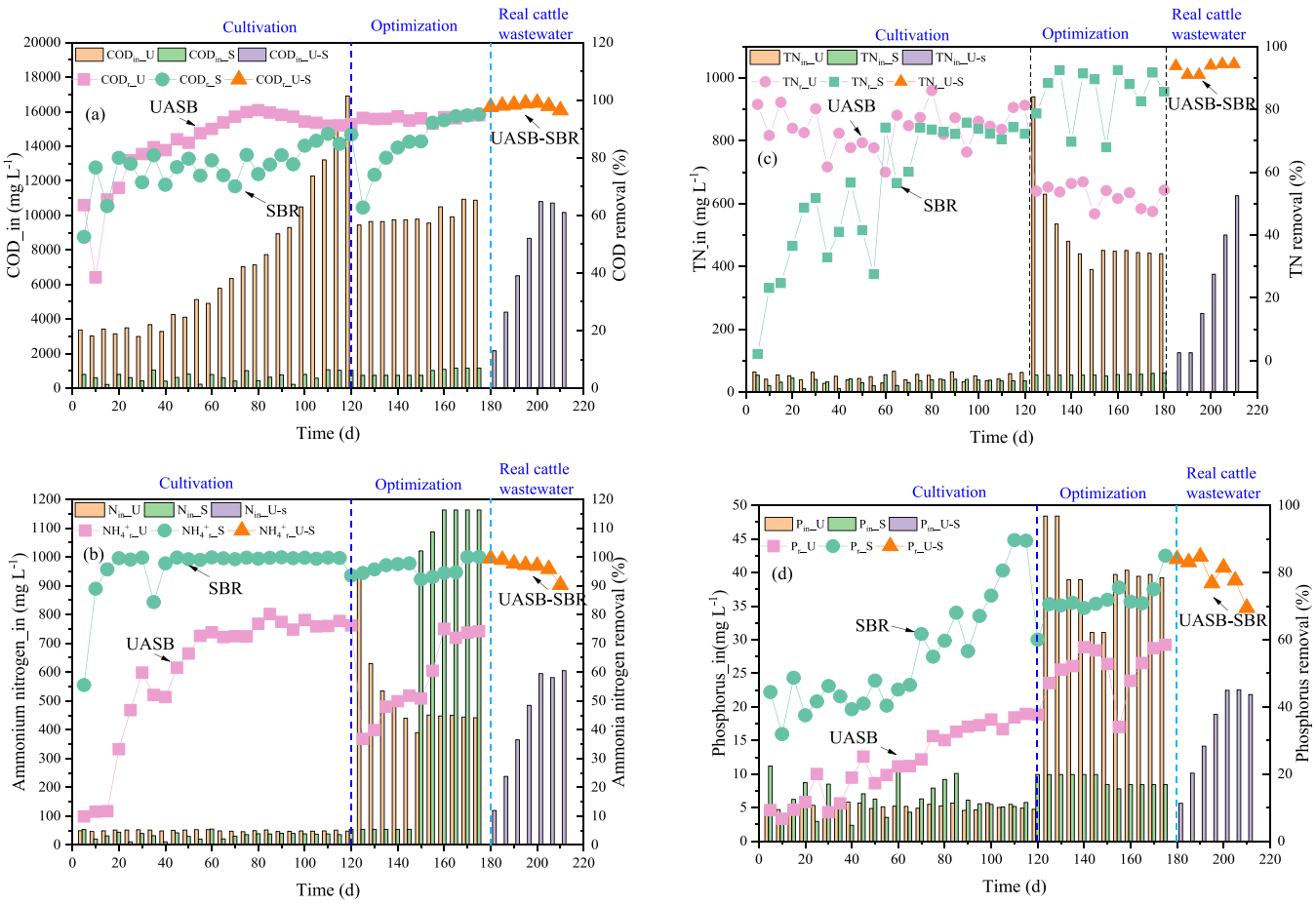



**Figure 2.** Concentration variation and removal efficiency of pollutants in UASB, SBR, and UASB−SBR: (a) COD, (b) $NH_4^+$-N, (c) TN, and (d) TP.

and fed them with real cattle wastewater. The properties of real cattle wastewater are summarized in Table 1.

**Chemical Analysis and Sludge Characterization.** Standard measurements were used to analyze COD, nitrogen ($NH_4^+$-N, $NO_3^-$-N, $NO_2^-$N, and TN), phosphorus (TP), and other items. The appearance of AGS under different conditions was characterized using a scanning electron microscope (SEM) from Hitachi Ltd., Japan (model: SU8020). Particle size was measured using a laser densitometer from Malvern Panalytical Ltd., Shanghai, China (model: Mastersizer 3000).

**Energy Balance Analysis.** Energy balance is presented with the energy balance ratio (EBR), which is the ratio of energy recovery converted from biogas produced by COD to energy consumption by the heater, aeration, and pumps in the throughput system.[17]

**Microbial Community Structural Analysis.** To analyze microbial samples, multiple portions of biofilms were collected using aseptic scissors under different conditions. Each condition was subjected to five repeated fed-batch cycles before the biofilm samples were collected. The microbial samples from the UASB (upflow anaerobic sludge blanket) reactor were designated as UASBi (inoculated activated sludge), UASB1 (anaerobic activated sludge in a stable state), UASB2 (anaerobic activated sludge under optimal conditions), and UASB3 (anaerobic activated sludge from real livestock wastewater). Similarly, the microbial samples from the SBR were named SBRi, SBR1, SBR2, and SBR3. Genomic DNA was extracted from the samples using the DNeasy PowerSoil Kit (Qiagen, Inc., Netherlands), as described in our published paper.[11] The pair of primers 338f (5′-ACT CCT ACG GGA GGC AGC A-3′) and 806r (5′-TCG GAC TAC HVG GGT WTC TAA T-3′) targeted the V3−V4 regions of the 16S rRNA genes, which were selected for polymerase chain reactions of amplicon library preparation.[18] Finally, the amplicons were sequenced using the Illumina MiSeq platform.

A total of 886,145 nucleotide sequence reads from eight samples were analyzed using QIIME 2.[19] The DADA2 algorithm was utilized for denoising, resulting in 417,278 high-quality paired-end reads with an average length of 251 bp.[20] These filtered reads were then clustered into 4183 amplicon sequence variants (ASVs), and taxonomy was assigned to them utilizing the Greengenes Databases (V13_8). Furthermore, a phylogenetic tree was generated using ARB software.[21] To predict the functional gene abundance, picrust2 was used based on the microbial community structure reflected by 16S rRNA gene sequencing data.[22] Moreover, canonical correspondence analysis (CCA) was conducted using the VEGAN package (V2.6-2) in R (version 4.0.3), based on the microbial community structure of genus level and the operation condition of the reactors.

## RESULTS

**UASB Performance.** The performances of single UASB, SBR, and integrated UASB−SBR were discussed according to the removal efficiencies of COD, $NH_4^+$, TN, and TP. The operating period was divided into three phases: phase I (0−

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120 days, cultivation period); phase II (121−180 days), optimization period; and phase III (181−220 days, real cattle wastewater fed), as displayed in Figure 2.

To investigate the effect of micro-aeration in UASB, two parallel UASB reactors were operated simultaneously. One of the reactors was equipped with top aeration, allowing a trace amount of DO in the reaction zone to simulate the effect of micro-aeration. The other reactor was operated under strictly anaerobic conditions. Initially, the COD removal efficiency of the UASB with micro-aeration was low and unstable (1−20 days, Figure 2a) due to the continuous washing out of sludge from the reactor (Supporting Information: Figure S1) resulting from the intake without anaerobic precipitation. Over the next 50 days, the anaerobic activated sludge in the UASB gradually transformed from loose flocs to small particles.

As the intake COD concentration rose from 3000 to 8000 mg $L^{-1}$, COD removal rates increased to 95% (micro-aeration) and 96% (strictly anaerobic). However, when the COD intake increased to 16,000 mg $L^{-1}$, the COD removal rates in both UASB reactors decreased to around 92% due to the occurrence of acidified influent inside the reactor, which inhibited the activity of methanogenic bacteria. This phenomenon indicates that excessive organic load negatively affects the role of microorganisms in COD removal. Despite the formation of AnGS in both reactors, the effluent COD concentrations were almost the same, when the sludge load was not too high, and the AnGS was adaptable.

When considering nitrogen removal in UASB reactors, we measured four parameters: $NH_4^+$-N, $NO_3^-$-N, $NO_2^-$-N, and TN. Initially, during the adaptation period, the concentration of $NH_4^+$-N remained relatively stable for the first 20 days as the anaerobic activated sludge was inactivated. However, as the anaerobic bacteria gradually adapted to the environment, differences between the two UASB reactors became more pronounced. Over the following 40 days, the $NH_4^+$-N removal rates increased to 70% in the top-aeration UASB and 30% in the strictly anaerobic UASB, respectively. Eventually, both UASB reactors reached a stable state by the 120th day, as depicted in Figure 2b.

Analyzing the nitrogen variations throughout the complete cycle, it was evident that the UASB system with top micro-aeration exhibited an impressive TN removal rate of 90%, with minimal accumulation of no more than 1% $NO_3^-$-N and negligible levels of $NO_2^-$-N. In contrast, the strictly anaerobic UASB achieved 75% of TN removal, including the accumulation of 23% $NO_3^-$-N and 2% $NO_2^-$-N (Supporting Information: Figure S2). These differences can be attributed to the fact that the top-aeration UASB reactor created a low DO condition, promoting the synergistic activity of facultative anaerobic, anaerobic, and aerobic bacteria, which led to higher nitrogen removal rates.

Regarding TP removal, both reactors experienced a fluctuating increase within the first 80 days and reached a relatively stable status of 40% removal by the 120th day. The top-aeration reactor exhibited a higher TP removal rate, which can be attributed to the enhanced biological activity facilitated by micro-aeration in the UASB reactor. The introduction of a low level of DO through micro-aeration promoted the activity of aerobic bacteria, which utilize oxygen for their metabolic processes. These aerobic bacteria played a crucial role in the bioconversion of phosphorus, resulting in higher phosphorus removal rates. Micro-aeration also fostered the growth of a diverse microbial community within the UASB reactor, comprising not only anaerobic bacteria but also facultative anaerobic and aerobic bacteria. The coexistence of these microbial groups created a synergistic effect, where their combined activities contributed to more efficient phosphorus removal. Additionally, the presence of oxygen in the micro-aerated UASB reactor facilitated the uptake of phosphorus by microorganisms, particularly aerobic bacteria, which have a higher affinity for phosphorus and can accumulate it within their cells. Consequently, the overall phosphorus removal efficiency of the system is enhanced. Furthermore, micro-aeration positively influenced the settling properties of the sludge in the UASB reactor. The presence of oxygen promoted the formation of denser and better-settling sludge flocs, improving settling characteristics and preventing the washout of biomass from the reactor. This ensured the retention of phosphorus-accumulating bacteria, further enhancing phosphorus removal efficiency.

The UASB reactor was optimized by adjusting the C/N ratios and HRT. Specifically, as the C/N ratio gradually decreased from 25:1 to 10:1, there was a slight drop in COD removal rate from 92 to 90%. However, when the HRT was extended from 12 to 48 h, the COD removal rate slightly increased from 92 to 94.5%. The COD trend indicates that the C/N ratio and HRT are not the sole limiting factors for simple substrates (Supporting Information: Figure S3). Following the optimization of the C/N ratio and HRT, the removal rates of $NH_4^+$, TN, and TP were recovered at 75, 99, and 60%, respectively (phase II in Figure 2). Therefore, a single UASB reactor can effectively remove organic matter but a part of nitrogen and phosphorus.

**SBR Performance.** The single SBR reactor experienced an adaption period (0−20 days) of which the removal rates of COD, $NH_4^+$-N, TN, and TP were relatively low. Then, the removal rates of these three parameters increased to 75, 60, and 20% on the 20th day, respectively, and continuously climbed and reached a stable status (88% COD, 75% TN, and 30% TP) on the 120th day, with a DO concentration of 3.0 mg $L^{-1}$ (phase I).

SBR was further optimized by varying the DO levels at 3, 4, 5, 6, 7, and 8 mg $L^{-1}$. The removal rate of COD increased to 95% when the DO level was enhanced to 6.0 mg $L^{-1}$. However, there was no further increase in the removal rate even when the DO level was increased to 8.0 mg $L^{-1}$ (Supporting Information: Figure S4). This trend indicates that the optimal DO level for the SBR is 6.0 mg $L^{-1}$ and increasing the DO level beyond that does not provide additional benefits. At a DO level of 6.0 mg $L^{-1}$, the SBR was optimized by varying the reaction times at 6, 8, 12, and 24 h. Only slight increases in the removal rates of COD were observed with an extended reaction time from 6 to 24 h, as shown in the additional figure (Supporting Information: Figure S4). This suggests that reaction time is not a crucial factor affecting pollutant removal and extending the reaction time consumes more energy. Under the conditions of a DO level of 6.0 mg $L^{-1}$ and a reaction time of 6 h, the SBR reactor was operated with two aeration modes, S1 and S2. The results indicate that the removal rates of COD and $NH_4^+$-N for S1 and S2 modes were not significantly different but the removal rates of TN and TP for S2 mode increased by 19 and 10.0%, respectively. These increases in the removal rates of TN and TP can be attributed to the alternating anaerobic and aerobic operation mode, which meets the requirements of enriching ammonia-oxidizing bacteria (AOB) and inhibiting nitrite-

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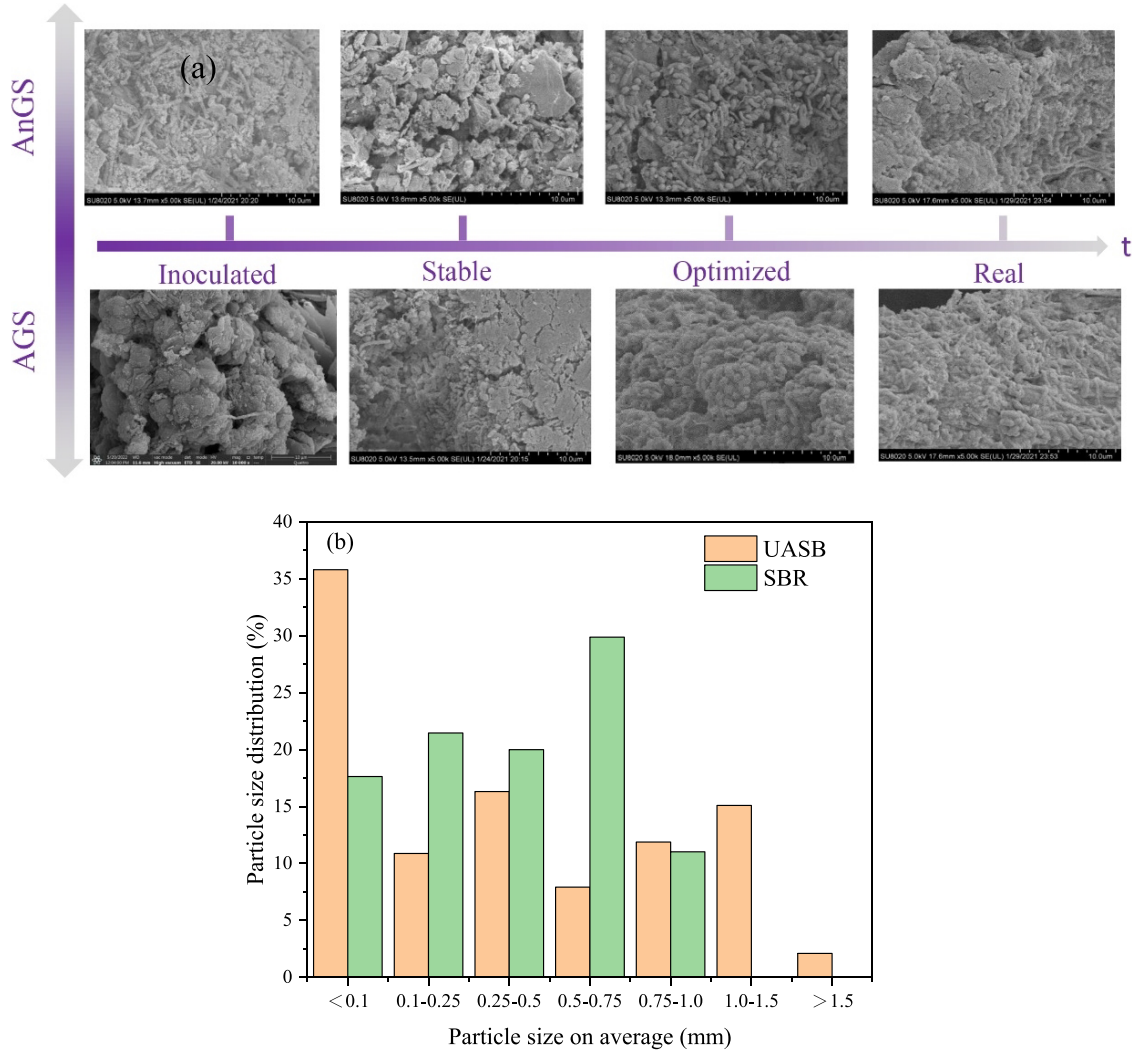



**Figure 3.** Activated sludge properties: (a) appearance of AnGS and AGS at the stages of inoculation, cultivation, optimization, and real cattle wastewater, and (b) average particle size of AnGS and AGS in UASB and SBR for real cattle wastewater.

oxidizing bacteria (NOB). This finding is consistent with the reported findings.[23] Finally, the SBR reactor was operated under optimal conditions (phase II) and the removal rates of COD, $NH_4^+$-N, TN, and TP were 94.0, 99, 94.4, and 95%, respectively, as displayed in Figure 2a–d.

**Real Cattle Wastewater Feeding.** In phase III, real cattle wastewater was diluted with synthetic wastewater at varying ratios of 20, 40, 60, 80, and 100% to mitigate the impact of concentrated pollutants. The UASB–SBR-integrated system was able to remove approximately 97% of COD, 95% of $NH_4^+$-N, 73% of TN, and 80% of TP when real cattle wastewater was at a concentration of 20%. As the proportion of real cattle wastewater increased to 100%, the rate of COD removal slightly decreased to 95% while the rate of $NH_4^+$-N, removal remained stable at 95% and the rate of TP removal dropped by 10%.

In addition, the use of intermittent aeration in the AGS process improved nitrogen removal efficiency, despite variations in objectives and conditions compared to the present study. Concentration profiles demonstrated the favorable impact of intermittent aeration on nitrogen removal, consistent with other studies.[24] These excellent results were attributed to the positive properties of the anaerobic/aerobic activated sludge and the functional microorganism community, as elaborated in the next section.

**Anaerobic/Aerobic Activate Sludge Properties.** Figure 3 illustrates the characteristics and particle distributions of anaerobic/aerobic activated sludge collected under different conditions. Initially, the inoculated sludge was composed of fine flocculent sludge, wherein the low activity and MLSS of the microorganisms prevented them from aggregating to form granular sludge. Subsequently, the flocculent-activated sludge grew with good settling performance and formed small particles with compact structures and regular shapes due to the interaction between microorganisms and hydraulic shear force. At the stable status (Figure 3a), AnGS particles became granules with diameters of about 1.0–1.5 mm, which were much larger than those observed in our previous study without micro-aeration.[11] Micro-aeration thus facilitated the self-agglomeration of microorganisms.

SEM was used to further characterize the morphological structures of mature AnGS. At a magnification of ×1000, the microscopic colonies consisted mainly of cocci and bacilli, interspersed with short filamentous bacteria. The long bacilli formed pore spaces on the surface inside the particles, which was important for biofilm formation and cell attachment to

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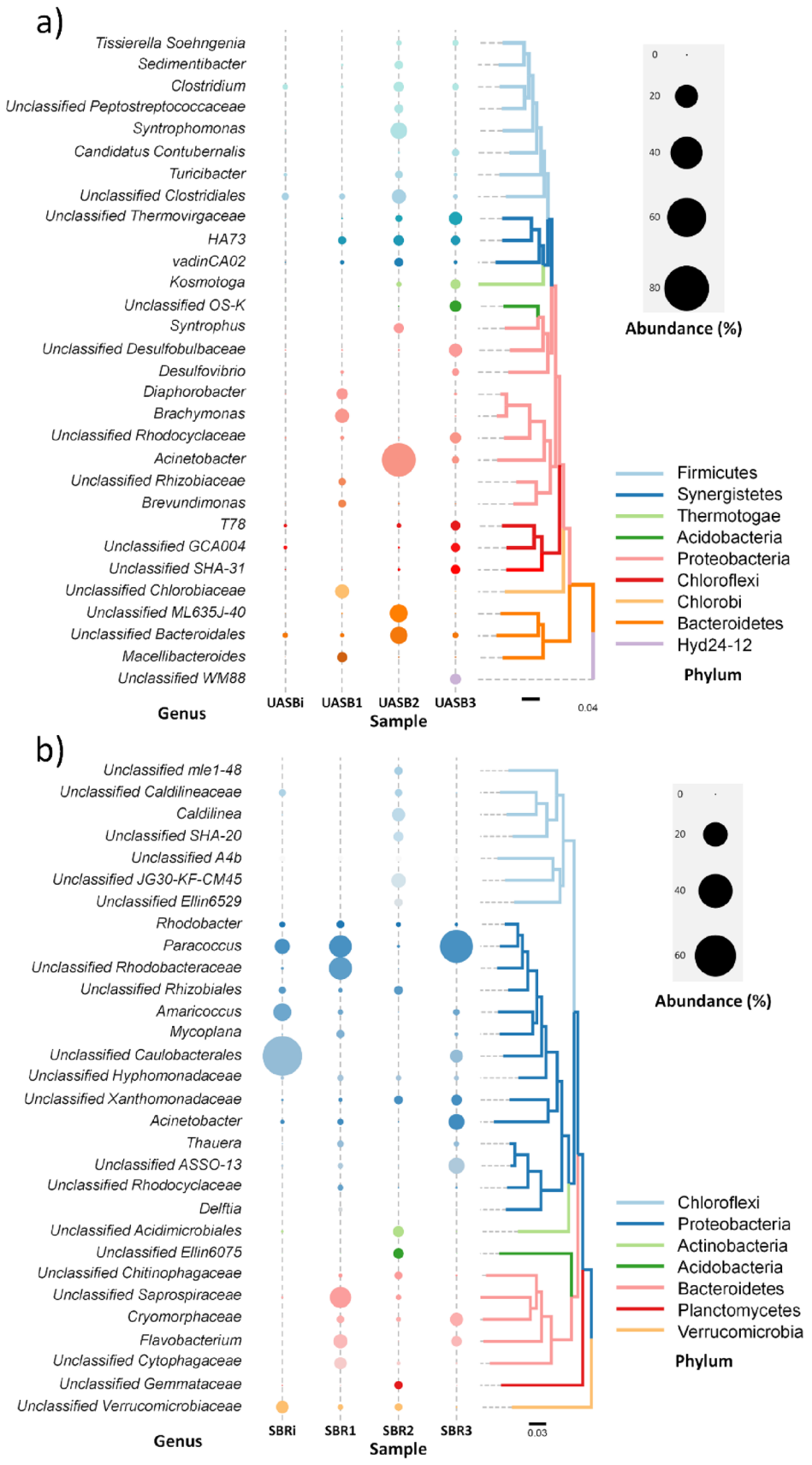



**Figure 4.** Microbial community structure in the phylum and genus levels in UASB and SBR reactors: (a) AnGS and (b) AGS.

surfaces.[25] When UASB was fed with real cattle wastewater, the microorganisms mainly comprised cocci and bacteria, which did not significantly change. The particle size distribution of AnGS is illustrated in Figure 3b. Anaerobic particles smaller



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than 0.1 mm accounted for 35% of the sample, while particles between 0.25 and 0.5 mm and between 1.0 and 1.5 mm accounted for 18% each. Particles larger than 1.5 mm made up only 2.5% of the sample. AnGS particles under optimal conditions were slightly smaller but much more compact than those in the stable state. When the influent was altered with real cattle wastewater, the color of AnGS in UASB changed from black to yellow-black and the size became 0.3−0.4 mm, slightly smaller than that under optimal conditions.

The characteristics of AGS are shown in the lower part of Figure 3a. Initially, the inoculated sludge consisted mainly of fine floc sludge and gradually developed into aerobic granules. Compared to the aerobic granules cultivated with S1 aeration mode, the aerobic granules cultivated with S2 aeration mode in this study were characterized by a larger size, a more rounded shape, and a smoother surface. The microorganisms, mainly consisting of cocci, were embedded in a non-filamentous extracellular matrix, which contributed to the regularly arranged and structurally dense AGS. The smooth surface indicates that the aerobic/anoxic cultivation mode is conducive to AGS formation. When real cattle wastewater was treated, the color of aerobic AGS turned gray and yellow and the microorganisms shifted to bacilli. The volume of aerobic AGS with a size of 0.5−0.75 mm was 30%, while those of 0.1−0.5 and 0.75−1.0 mm were 45 and 12%, respectively. Intensive intermediate aeration contributed to better settling performance, resulting in the formation of AGS with a denser structure and a more regular shape than continuous and long aeration. The changes in granular sludge demonstrate that intermittent aeration inhibits the growth of filamentous bacteria and accelerates the granulation process of activated sludge.

**Microbiological Community and Structure in UASB.** The microbiological composition of the UASB reactor was analyzed, and the top 30 genera in relative abundance are shown in Figure 4a. These genera belonged to nine phyla, namely, *Firmicutes*, *Synergistetes*, *Thermotogae*, *Acidobacteria*, *Proteobacteria*, *Chloroflexi*, *Chlorobi*, *Bacteroidetes*, and *Hyd24-12*. It was observed that the microbial community structure in the UASB reactor was significantly different from that of the inoculated sludge (UASB1).

During the initial stage, the most predominant genus (UASBi) was *Acinetobacter*, which belonged to *Proteobacteria* and has a relative abundance of 46.52%. *Acinetobacter* is a group of non-fermentative aerobic bacteria that are widely distributed in wastewater treatment systems. They have the capability to degrade non-glucose carbon sources, such as short-chain fatty acids, phenol, and alcohol.[26] Other genera that had comparatively high relative abundance included *Brachymonas* (7.91%) and *Diaphorobacter* (5.09%) from *Proteobacteria*, *Macellibacteroides* (4.06%) from *Bacteroidetes*, and *Chlorobiaceae* (7.87%) from *Chlorobi*. All of these genera can ferment carbohydrates into diverse organic acids.[27] *Brachymonas* bacteria serve a structural function and contribute to the formation of sludge granules in wastewater treatment systems.[28] *Brachymonas* and *Diaphorobacter* have also been reported as denitrifying bacteria in granular sludge.[29,30] *Macellibacteroides* plays a significant role in polymer degradation and the production of volatile fatty acids (VFAs).[31] Furthermore, *Chlorobiaceae* has been identified as green sulfur bacteria that utilize sulfur, sulfide, and thiosulfate as electron donors.[32]

In phase II, the abundance of *Proteobacteria* in the granular sludge (UASB2) decreased and was replaced mainly by microorganisms from *Firmicutes*, *Synergistetes*, and *Bacteroidetes*. Among these phyla, the highest abundance was found in *ML635J-40* (12.87%) and *Bacteroidales* (11.92%) from *Bacteroidetes*, *Syntrophomonas* (11.05%), and *Clostridiales* (8.09%) from *Firmicutes*. *Bacteroidetes*, represented by *Bacteroidales*, exhibit saccharolytic and proteolytic activity in anaerobic systems, while some short-chain fatty acids such as propionate, acetate, and succinate are produced.[33] *Clostridiales* mainly hydrolyze and ferment carbohydrates.[34] Bacteria in *Syntrophomonas* degrade long-chain fatty acids via *β*-oxidation.[35] Members of the *Synergistetes* phylum play a prominent role in protein and amino acid degradation.[36] Meanwhile, *Clostridiales* and *Syntrophomonas* utilize short-chain fatty acids through syntrophic metabolism with methanogens and supply hydrogen and acetic acid for methane production.[37]

When real wastewater (UASB3) was introduced to the UASB reactor (phase III), the abundance of bacteria in several phyla, including *Synergistetes*, *Thermotogae*, *Acidobacteria*, *Proteobacteria*, *Chloroflexi*, and *Hyd24-12*, significantly increased. Conversely, the bacterial abundance of *Bacteroidetes* and *Firmicutes* decreased. Notably, the increase in sulfur-reducing bacteria such as *Unclassified ASSO-13* (6.71%) and *Desulfovibrio* (1.92%) from *Proteobacteria*, as well as *Acinetobacter* (2.06%) from *Acidobacteria*, may be linked to the presence of sulfur-containing proteins in the real poultry wastewater.[38] Moreover, the abundance of *Synergistetes*, a group of protein-metabolizing bacteria, also increased. Other microorganisms showed an increase due to the presence of complex organic matter and other substances in the real poultry wastewater. The *Anaerolineae* group (*Unclassified GCA004*, 3.60%; *Unclassified SHA-31*, 3.27%; and *Unclassified Chlorobiaceae*, 0.02%), known as fatty acid-oxidizing syntrophs in the processes,[39] showed an increase. *Kosmotoga* (3.82%) was also reported to be associated with the degradation of some complex organic compounds.[40]

**Microbiological Community and Structure in SBR.** In Figure 4b, the microbial composition of AGS in an SBR reactor is presented. The top 30 genera, in terms of relative abundance, belonged to seven phyla: *Chloroflexi*, *Proteobacteria*, *Actinobacteria*, *Acidobacteria*, *Bacteroidetes*, *Planctomycetes*, and *Verrucomicrobia*, respectively. Among these, the *Unclassified Caulobacterales* genus from *Chloroflexi* was the most prevalent in the inoculum, with a relative abundance of 57.40%. However, it was gradually replaced by the dominant genera *Paracoccus* and *Unclassified Rhodobacteraceae*, belonging to *Proteobacteria*, whose corresponding relative abundances at a stable status were 73.88 and 61.14%, respectively. *Paracoccus* sp. is known for its versatile metabolism and plays a significant role in biogeochemical cycles. This genus has been successfully employed in the biotreatment of contaminated environments, and one complete genome of *Paracoccus denitrificans* PD1222 has been deposited in the NCBI database.[41] The *Paracoccus denitrificans* narK gene encodes a transmembrane protein with a NarK1 domain that functions as a $NO_3^-/H^+$ symporter to initiate respiration and a NarK2 domain that acts as a $NO_3^-/NO_2^-$ antiporter.[42] This genus can rapidly reduce nitrate and nitrite and is capable of growing aerobically and anaerobically when either of these compounds is present in the medium.[43]

After the DO and HRT optimizations, the microorganisms in SBR exhibited increased gene diversity, and the dominant genus included *Unclassified Ellin6075* (3.59%) from *Acidobacteria*, *Unclassified Acidimicrobiales* (4.16%) from *Actino-*

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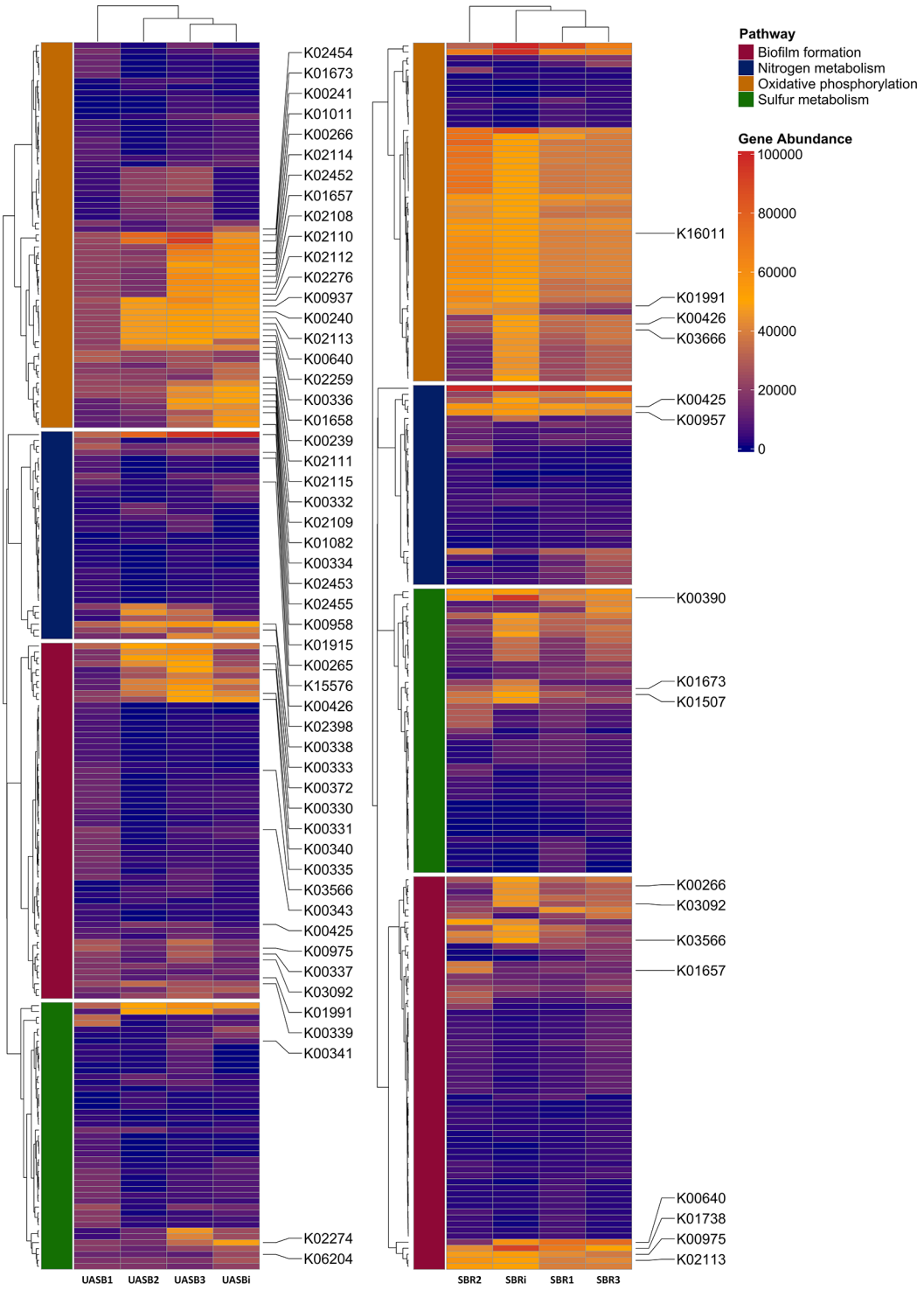



**Figure 5.** Abundance of functional genes involved in granular formation, oxidative phosphorylation, nitrogen metabolism, and sulfur metabolism.

*bacteria*, *Unclassified Gemmataceae* (2.16%) from *Planctomycetes*, *Unclassified Ellin6529* (2.19%), *Unclassified JG30-KF-CM45* (7.38%), *Unclassified SHA-20* (3.35%), *Caldilinea* (6.10%), *Unclassified Caldilineaceae* (1.74%), and *Unclassified mle1−4* (2.16%) from *Chloroflexi*. The relative increase in abundance could be attributed to the rise in the percentage of DO.[44] The family *Gemmataceae* thrives in aerobic conditions,[45] while *Caldilineaceae*, which was highly represented, has been identified as phosphorus-accumulating organisms in wastewater treatment systems.[46]

When the substrate was changed to real cattle wastewater, *Paracoccus* was found to be the most predominant genus, with a relative abundance of 39.80%, which is related to the denitrification process. The high percentage of members belonging to *Proteobacteria* is consistent with previous studies that evaluated the ammonia tolerance of microbial communities in activated and granular sludge. *Unclassified* ASSO-13 (9.10%) from *Chloroflexi* and *Acinetobacter* (9.04%) from *Proteobacteria* were both found to be important for enhanced biological phosphate removal from wastewaters.[47] The differ-

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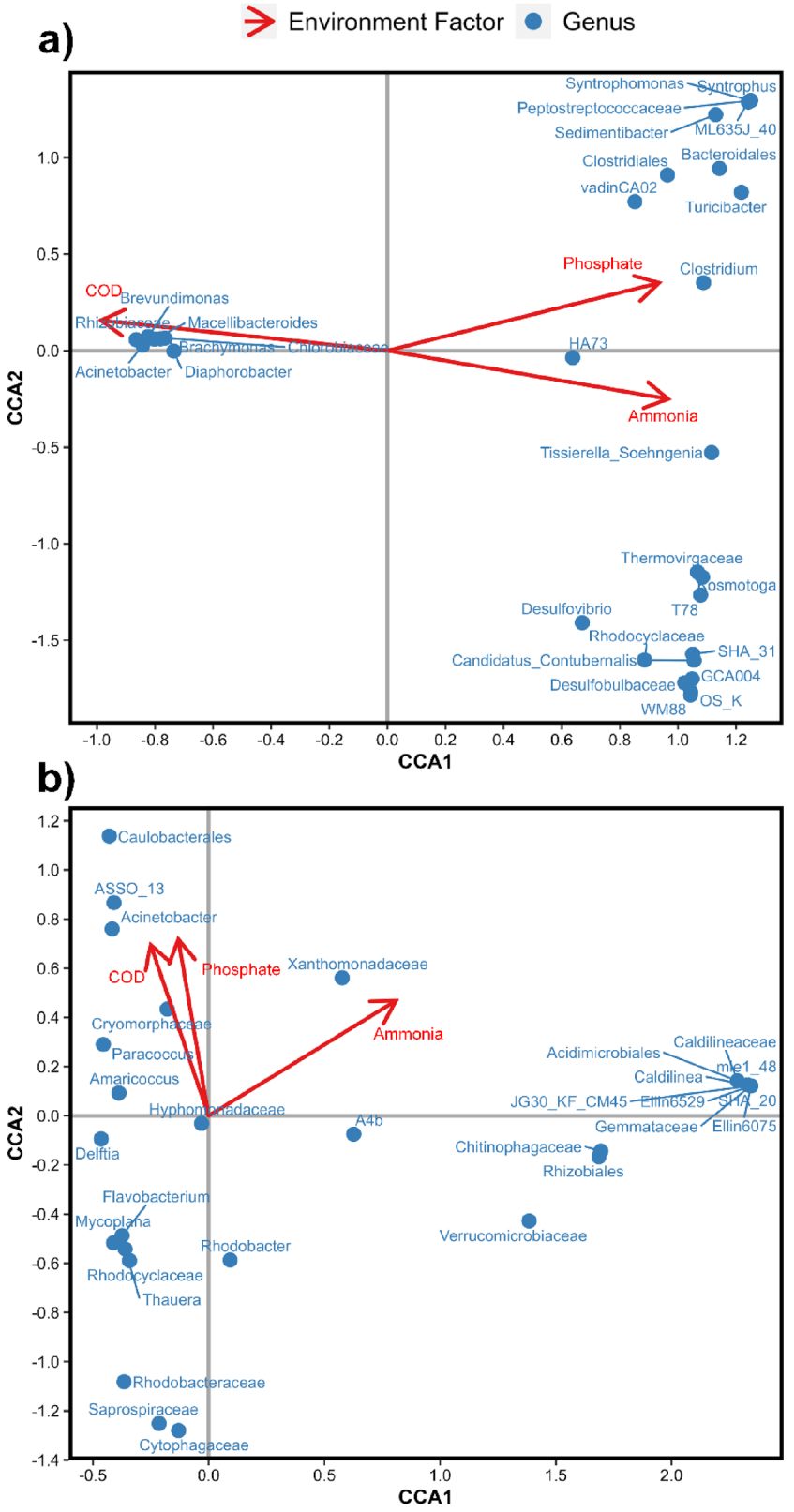



**Figure 6.** Canonical correspondence analysis of microorganisms: (a) UASB and (b) SBR at the genus level versus characteristics of influent (ammonia, phosphate, and COD).

ences in relative abundance of genera between synthetic and real cattle wastewater suggest that *Paracoccus* has better adaptability to the environment than *Acinetobacter*. Therefore, the complex composition of cattle wastewater alters the microbial living environment.

**Functional Gene Prediction.** To gain a deeper understanding of the specific roles played by microbial communities

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in granular sludge, we employed PICRUSt2 to predict gene abundance based on the results of 16S rRNA gene amplicon analysis. Subsequent analysis of the KEGG Pathway Database revealed the involvement of various enzymes encoded by corresponding functional genes in processes such as granular formation (red), oxidative phosphorylation (orange), nitrogen metabolism (purple), and sulfur metabolism (green), as summarized in Figure 5. Notably, AGS in the SBR reactor exhibited a more intense oxidative phosphorylation process, which was attributed to the microorganisms' energy acquisition mechanisms in the two reactors. These conditions were favorable for the survival of nitrogen removal flora. In contrast, AnGS in the UASB reactor showed higher levels of oxidative phosphorylation genes during the optimal and cattle wastewater feeding stages. Anaerobic respiration, similar to aerobic respiration, was beneficial because it was more efficient than fermentation and allowed for ATP synthesis via electron transport and oxidative phosphorylation in the absence of oxygen.[48]

We conducted an analysis of the functional gene involved in nitrogen metabolism and found that nitrogen metabolism was significantly enriched, which is consistent with the abundances of denitrifiers detected in the SBR reactor during different phases. The predicted functional genes involved in nitrogen cycling, such as nitrification, assimilatory and dissimilatory nitrate reduction, denitrification, and nitrogen fixation, are illustrated in Figure 5. Genes encoding assimilatory and dissimilatory nitrate reduction and denitrification, as well as multifunctional genes encoding nitrate reductase/nitrite oxidoreductase, were found to have a high abundance, demonstrating the high nitrogen metabolism potential of the denitrifying sludge in the SBR.[49] However, the abundance of critical genes involving nitrogen removal slightly decreased due to the altered circumstances caused by the introduction of real cattle wastewater. These genes encoded enzymes such as nitrite oxidase, nitrite reductase, nitrate reductase, nitric oxide reductase, nitrous oxide reductase, assimilatory nitrate reductase, and dissimilatory nitrate reductase.[50] For the anaerobic sludge, the important genes involved in nitrogen degradation are also shown in Figure 5, but they differ from those in the SBR. The encoded glutamate dehydrogenase is involved in the metabolism of glutamate to 2-oxoglutarate.[51] This enzyme releases ammonia in a reversible $NAD(P)^+$-dependent oxidative deamination of glutamate that yielded 2-oxoglutarate (2OG).[52]

Granules can be considered a special type of biofilm, and as such, the abundance of genes related to biofilm formation was predicted.[53] The encoded polysaccharide biosynthesis/export protein (K01991) and glycine cleavage system transcriptional activator (K03566) in AnGS for all samples were higher than in other samples. These two genes are involved in cell growth, specifically in the biosynthesis of bacterial polysaccharides, which is a complex process that requires membrane and soluble enzymes, as well as a transmembrane translocation step.[54] The abundance of polysaccharide biosynthesis/export protein and glycine cleavage system transcriptional activator varied with the cultivation conditions, particularly during the optimization stage when the abundance of the gene coding for polysaccharide biosynthesis/export protein changed. Notably, the abundance of the gene related to cell biosynthesis was restored when SBR was fed on real cattle wastewater. In UASB, the genes with high abundance were involved in VFA generation in AnGS, such as K01657, K00975, and K03092, which represent anthranilate synthase component, glucose-1-phosphate adenylyltransferase, and RNA polymerase, respectively.[55−58] The comparison of gene abundance among UASBi, UASB1, UASB2, and UASB3 indicates that bacteria become more active and synthesize more extracellular polymeric substances (EPS) to adapt to the complex substrates present in real cattle wastewater. Furthermore, the gene abundance involved in sulfur metabolism differed in SBR compared to UASB. The functional genera revealed that the dominant denitrifying bacteria and sulfate-reducing bacteria (SRB) enriched the granular sludge in both SBR and UASB.

**CCA.** To understand how different microorganisms in a reactor are influenced by various influent characteristics, we performed CCA on the top 30 microorganisms at the genus level, ranked by their relative abundance, in UASB and SBR. We examined the influent concentrations of ammonia, COD, and phosphate. As shown in Figure 6a, the effect of COD and ammonia/phosphate on microorganisms in UASB was reversed. Certain microorganisms in the UASB were more affected by the concentration of COD (such as *Acinetobacter*, Unclassified *Bacteroidales*, and *Brevundimonas*), which was confirmed by the high removals under optimal conditions (see UASB Performance). Genus *HA73* and Clostridium were found to be sensitive to ammonia concentration, showing positive correlations with ammonia. Genus *HA73* degrades VFAs to intermediate substances, such as acetate, hydrogen, and carbon dioxide for methanogens.[59] However, the family of HA73 (*Synergistetes*) was vulnerable to other environmental conditions, such as ammonia concentration, temperature, and retention time.[60] Members of Clostridium participate in acetogenesis via the production of extracellular enzymes for degrading biopolymers to substrates for CH4 production under strictly anaerobic conditions.[61] Genus *Clostridium* was involved in phosphate-dependent gene regulation, and its growth was greatly inhibited by phosphorus accumulation.[62] Another group of microorganisms, including *Syntrophus*, *Syntrophomonas*, *Clostridiales*, and *Bacteroidales*, was seen to be more affected by the phosphate concentration.

The effects of COD and phosphate on microorganisms in the SBR reactor were found to be more consistent, while the effects of ammonia nitrogen were relatively independent (Figure 6b). Genera such as *Acinetobacter*, Unclassified *Cryomorphaceae*, *Caulobacterales*, and *ASSO_13* were primarily affected by COD and phosphate concentration. The functional groups of Acinetobacter could potentially play a crucial role in pollutant reduction in the system, such as denitrifying phosphate-accumulating organisms.[63] Additionally, the relative abundance of Acinetobacter in real cattle wastewater (SBR 3) was observed to be higher than that in synthetic wastewater (SBR2) under optimal conditions. *Cryomorphaceae*, a family of bacteria in the order *Flavobacteriales*, were responsible for EPS production. The growth of numerous polyphosphate-accumulating organisms (PAOs) belonging to the taxa Cryomorphaceae was found to be distributed in the surface layer and inner layer of granular sludge.[64] *Cryomorphaceae* played an essential role in the carbon and energy cycle of extreme environments, and most bacteria of this family were found in locations rich in organic carbon but grew relatively slowly, with colonies taking at least 7 days to appear and grow to full size.[65] In SBR, the abundance of unclassified *Xanthomonadaceae* showed a positive correlation with ammonia. A possible explanation could be heterotrophic nitrification mediated by ammonia-oxidizing

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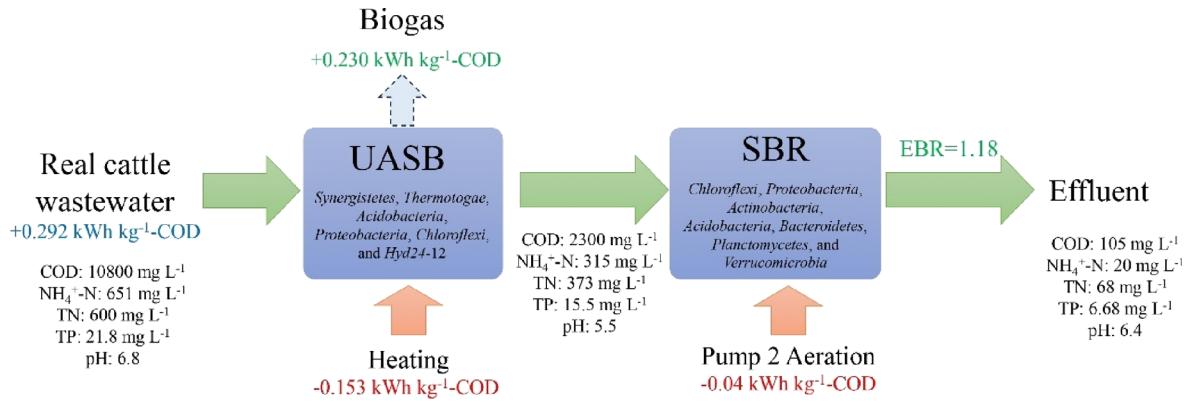



**Figure 7.** Energy flow in the UASB−SBR-coupled system.

**Table 2. Performance Comparison among Physico-Chemical Approaches, Phytoremediation, and Biological Approaches**[a]

| | | SS | turbidity | COD | $NH_3$-N | $PO_4^{3-}$P | energy con. | energy reco. | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | processes | removal rate (%) | | | | | kWh $kg^{-1}$ COD | | EBR | ref. |
| physico-chemical approaches | multichannel ceramic ultrafiltration membranes | 100.00 | 99.20 | 19.72 | 3.12 | 80.21 | N.M | N.M | N/A | 69 |
| | modified zeolite and a bioflocculant | N.M | 95.5 | 88.6 | 85.8 | N.M | N.M | N.M | N/A | 70 |
| | advanced oxidation | 100 | N.M | 83 | 100 | 100 | N.M | N.M | N/A | 71 |
| | electrochemical oxidation | 99.5 | N.M | 92.2 | 93.5 | 100 | 0.154 | N.M | N/A | 72 |
| | precipitation with ultrasonic pre-treatment | N.M | N.M | N.M | 20 | 80 | 0.278 | N.M | N/A | 73 |
| | coagulation and flocculation | 87.8 | N.M | 67.3 | 96.2 | 96.9 | N.M | N.M | N/A | 74 |
| | electrodialysis | N.M | N.M | N.M | 100 | 84 | 0.13 | N.M | N/A | 75 |
| | photocatalytic system (UV-$TiO_2$) | 88 | N.M | 66 | 60 | 22 | N.M | N.M | N/A | 76 |
| phytoremediation | microalgae | N.M | N.M | 62.3 | 81.16 | 85.29 | N.M | N.M | N/A | 4 |
| | aquatic plants | N.M | N.M | 93.7 | 66.4 | N.M | N.M | N.M | N/A | 77 |
| | constructed wetlands | N.M | N.M | 77 | 71.5 | 62 | N.M | N.M | N/A | 78 |
| biological approaches | sequencing batch reactor | 61.4 | N.M | 80.3 | 31.2 | 87.2 | N.M | N.M | N/A | 79 |
| | anaerobic digestion | 92 | N.M | 78 | N.M | N.M | 0.07 | 0.252 | 3.59 | 80 |
| | UASB−SBR-integrated system | N.M | N.M | 96.93 | 99.47 | 90 | 0.195 | 0.230 | 1.18 | this study |

[a]Note: N.M, not measurement; N/A, not available.

*Xanthomonadaceae* in SBRs operated with real cattle wastewater.[66]

**Energy Balance Analysis.** The energy balance of the UASB−SBR system was estimated to determine its net energy efficiency. Energy consumption occurred through the use of heaters, pumps, and aeration, while energy recovery was achieved through biogas production. As shown in Figure 7, the total energy stored in real cattle wastewater was approximately 0.292 kWh $kg^{-1}$-COD. The energy consumption of the UASB and SBR processes was 0.153 and 0.04 kWh per kg of COD, respectively. The residual energy stored in COD was 0.06 kWh per kg of COD, and the recovered biogas was 0.230 kWh per kg of COD.[17] The final EBR was 1.18, indicating that the system achieved a net energy efficiency of 18% through the UASB−SBR coupling. However, further investigation revealed that the energy consumption of the UASB and SBR processes was primarily due to the use of pumps, aeration, and heating. Specifically, the pumps of UASB and SBR consumed 0.012 and 0.016 kWh per kg of COD, respectively, while the aeration process used 0.023 kWh per kg of COD. The heating process consumed the most energy, accounting for almost 70% of the total energy consumption, and was necessary to maintain high activities of hydrolytic and methanogenic bacteria. The heating energy consumption was 0.14 kWh per kg of COD.

Table 2 presents a summary of published studies on cattle wastewater treatment, focusing on aspects such as suspended solids (SS), turbidity, COD, $NH_3$-N, $PO_4^{3-}$-P, and energy balance. The studies were divided into three categories: physico-chemical, phytoremediation, and biological oxidation. Physico-chemical techniques, such as multichannel ceramic ultrafiltration membranes, modified zeolite and a flocculant, advanced oxidation, electrochemical oxidation, precipitation with ultrasonic pre-treatment, coagulation and flocculation, electrodialysis, and photocatalytic systems (UV-$TiO_2$), have been employed for cattle wastewater treatment. Physico-chemical processes demonstrated high removal efficiencies of SS, turbidity, COD, $NH_3$-N, and $PO_4^{3-}$-P. However, they did not consider energy consumption. For example, multichannel ceramic ultrafiltration membranes can remove 100% of SS, 99.2% of turbidity, and 80.21% of $PO_4^{3-}$-P. Others obtained high removal efficiencies at the cost of intensive energy consumption. For instance, electrochemical oxidation can remove 99.5% of SS, 92.2% of COD, 93.5% of $NH_3$-N, and 100% of $PO_4^{3-}$-P, but at a consumption of 0.154 kWh $kg^{-1}$-COD.

In this study, a UASB−SBR-integrated system removed 96.95% of COD, 99.47% of $NH_3$-N, and 90% of $PO_4^{3-}$-P from real cattle wastewater. Most importantly, this system yielded an



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18% net energy output, indicating the significant potential of cattle wastewater. However, the potential of the UASB−SBR-integrated system was underestimated due to the issue of low activity of low-temperature methanogens. Thus, future improvement measures could include isolating methanogenic colonies exposed to low-temperature environments in alpine areas and inoculating these colonies (cold-adapted colonies) into the UASB reactor, employing bio-electrochemical systems to maintain methanogenic activity at low temperatures,[67] or enhancing direct interspecies electron transfer by using exogenous non-biological conductive materials such as granular activated carbon, biochar, and carbon nanotubes.[68]

## CONCLUSIONS

This study developed a coupled system consisting of top-aired UASB and SBR to simultaneously recover biogas and remove nutrients from cattle wastewater. Prior to the coupling of UASB and SBR, each component was independently optimized. The results demonstrated that UASB operated with top micro-aeration had larger AnGS particles compared to that operated under strict anaerobic conditions. Similarly, the SBR operated with S2 aeration mode produced larger AGS particles with round shapes and smooth surfaces when compared to the control group (S1). Under the optimal conditions, this hybrid system contributed 94% of COD, 90% of $NH_4^+$-N, 90% TN, and 61% of $PO_4^3$-P removed from real cattle wastewater, with the cooperation of *Synergistetes*, *Thermotogae*, *Acidobacteria*, *Proteobacteria*, *Chloroflexi*, and *Hyd24-12* dominated in UASB, and *Paracoccus*, *Proteobacteria*, *Chloroflexi*, and *Acinetobacter*. The energy balance analysis revealed that the generated biogas accounted for 18% of the net energy. These findings highlight the potential of the UASB−SBR-coupled system as an effective approach for energy recovery from cattle wastewater.

## ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsestwater.3c00241.

Typical pollutant removal performance of UASB without micro-aeration (Figure S1); analysis of nitrogen composition variations (Figure S2); optimization of organic load, C/N ratio, and HRT (Figure S3); analysis of SBR performance with different DO and reaction times (Figure S4) (PDF)

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### Author Contributions

#L.X. and J.H. contributed equally to the study. CRediT: **Jiansheng Huang** funding acquisition (equal); **Yinying Guo** methodology (equal).

### Author Contributions

L.X. and J.H. equally conducted experiments and drafted and revised this manuscript. Y.G. and R.Z. verified the analytical methods. L.L., H.Z., and Q.H. conceived of the presented idea and supervised the findings of this work. S.-Y.L. revised this manuscript. All authors have given approval to the final version of the manuscript. CRediT: **Jiansheng Huang** funding acquisition (equal); **Yinying Guo** methodology (equal).

### Notes

The authors declare no competing financial interest.

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